

Research Article

Study on the Vortex-Induced Vibration Performance of the LSOB Girder of a Valley-Crossing Suspensions Bridge

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The lower semi-open box (LSOB) girder is widely used in large-span bridges due to its favorable stress performance and cost-effectiveness. The vortex-induced vibration (VIV) of the LSOB girder was studied by using a 1:55 section model wind tunnel test. The test results show that there are obvious vertical and torsional VIV intervals with significant amplitudes at each angle of attack with the damping ratios of about 0.35%. The CFD numerical simulations indicate that VIVs of LSOB girders are mainly induced by two types of vortices: one at the inclined webs of leeward wind nozzles and another at the upstream upper surfaces of the deck. Conventional aerodynamic vibration suppression measures, such as the lower central stabilizing plate and the guide vane at the gantry rail, demonstrate that the lower central stabilizing plate can reduce the VIV response at all wind angles of attack, decreasing vertical amplitude by 40%–50% and torsional amplitude by 10%–25%. Although adding a guide vane further optimizes the VIV performance, its effect is limited. Installing the lower central stabilizing plate helps redistribute vortices below the girder, transforming large vortices into smaller ones, most of which do not detach, thereby reducing vortex shedding intensity in the wake area. The bilateral guide vanes at the maintenance car track further weaken the wake vortices at the diagonal webs on the leeward side.

Keywords: aerodynamic countermeasure; CFD; dynamic mesh technology; LSOB girder; vortex-induced vibration (VIV); wind tunnel test

1. Introduction

Steel box girders have always been one of the main types of girders used in large-span bridges, characterized by their lightweight, high cross-sectional stiffness, and good stability. A new girder type with lighter weight and more advantageous manufacturing cost has emerged, the lower semi-open box (LSOB) girder, based on the traditional monolithic steel box girder. This type reduces steel usage and enhances structural stiffness by eliminating the lower deck and increasing stiffness at the wind nozzles, thus improving stability. LSOB girders are similar to π -type girders in that they are both open girder types relying on the side girders as the main load-bearing members, both of which have the

advantages of low cost, good load-bearing performance, and easy maintenance. However, compared with the traditional π -type girder, the LSOB girder uses box beams on both sides as the side girder, so its stiffness is greater than that of the I-type side beams in the π -type girder. In bridges that require large stiffness of main girders, LSOB girders have the advantage of high stiffness.

Vortex-induced vibration (VIV) is a self-excited and amplitude-limited oscillation which is caused by the shedding vortices. Although the VIV is unlikely to cause direct damage and significant amplitude vibration for large-span bridges, it inevitably affects the comfort and safety of pedestrians and drivers. Thus, VIVs should be mitigated according to many guidelines of construction codes [1, 2].

And, the installation of aerodynamic countermeasures is an effective method to improve the VIV performance by tailoring the deck shape [3–5].

A large number of researchers have studied the VIV phenomenon and proposed various aerodynamic countermeasures to reduce or even eliminate the VIV. Reference [6] conducted a series of numerical simulations to clarify the shaping effect of triangular wind fairings on the aerodynamic behavior of a rectangular bridge deck. It is found that the bottom plate slopes of 25° – 15° of triangular wind fairing have lower aerodynamic responses. Reference [7] conducted a series of wind tunnel tests to investigate the VIV performance of a rectangular steel box girder and proposed a quadrilateral wind fairing with a platform to eliminate the VIV through the CFD method. And, the suppression efficiency is verified by wind tunnel tests. Reference [8] studied the mechanism of a multiple-order vertical VIV of a typical streamlined box girder of a famous suspension bridge in China which was observed to be suffering from VIVs in the spring of 2020. In addition, Reference [9] conducted some numerical simulations to investigate the influence of the water-filled barrier on the VIV performance of the same bridge. Reference [10] utilized the wind tunnel test method to study the suppression efficiency of the moveable guide vanes on the VIV of a streamlined box girder. The results showed that the new type of removable guide vane, installed at the maintenance pedestrians, can effectively suppress the box girder's VIVs. Reference [11] proposed an experimental data-driven adaptive surrogate-based optimization approach which can be used in the optimization of deck shapes to improve the VIV performance of twin box girders. Reference [12] proposed an active jet device generated by an airflow velocity regulator installed on a box girder, which can disrupt the strong alternating vortex motion and shedding in the wake. Thus, the periodic aerodynamic forces exerted by the airflow could be reduced. It was considered that the proposed countermeasure is more effective than the fixed aerodynamic measurement on the suppressing VIV. Reference [13] developed a wind tunnel test setup for a sea-crossing bridge to investigate and improve the VIV performance of a streamlined box girder by using a large-scale model test and proposed a combined aerodynamic horizontal stabilizer. The width of the stabilizer can be adjusted to fit the bridge. The results of systematic wind tunnel tests show that the shape of the wind fairing has a significant effect on the VIV response of blunt box girders, which was verified by References [7, 14–17].

Meanwhile, many studies have shown that, compared with the streamlined steel box girder section, the semi-enclosed box girder removes part of the bottom plate and replaces it with a transverse diaphragm distributed along the longitudinal direction of the bridge. An enormous vortex will be formed at the half-opening position, which affects the flow-around structure and thus weakens the vortex-excited vibration performance of the girder. Reference [18] conducted a series of wind tunnel tests and found the measurements to improve the VIV performance of a π shape girder. Reference [16] developed a new empirical model of

nonlinear self-excited force and studied the mechanism of the torsional VIV of a π shape girder.

For the π -girder section, which has the same opening characteristics as the LSOB girder, it has been shown that the VIV performance of the π shape girder can be effectively improved by a suitable barriers closure scheme, which was verified by References [18, 19]. Many research results also show that installing lower stabilizer plates can effectively improve the vortex shedding at the lower open section of π shape girders. References [20–22] proposed some measurements, such as skirt shape plate and wind nozzles, to investigate the suppression of a vortex-excited vibration of π shape girders.

The aforementioned investigations have demonstrated that different girder types exhibit dramatically varying suppression effects from similar aerodynamic methods. There is a dearth of basic research on the impacts of vibration suppression measures, and there is a lack of information regarding the features of eddy vibration and pneumatic vibration suppression measures for LSOB girders. Therefore, this paper focuses on an LSOB girder double-tower suspension bridge with a main span of 880 m, first using a 1:55 scale sectional model wind tunnel test to examine the VIV characteristics of the LSOB main girder in detail. Concurrently, dynamic mesh and static CFD numerical simulations are employed to compare and analyze the LSOB girder's aerodynamic characteristics and the VIV's triggering mechanism. Subsequently, by combining wind tunnel tests and CFD numerical simulations, the influence and mechanism of two standard aerodynamic measures—namely, the lower central stabilizing plate and the double-side guide vanes at the maintenance car track—on the VIV performance of the LSOB girder are investigated in detail. This study aims to provide a basis and reference for the future design of the LSOB girder profile.

2. VIV Performance of the Original LSOB Girder

In this study, a double-tower suspension bridge which crosses a valley in Yunnan province serves as the engineering background, with a span arrangement of 305 + 880 + 318 m and a rise-span ratio of 1/8.8. As depicted in Figure 1, the height of the bridge tower on one side of the 318 m-long side span is 110.2 m, and the value of the tower on the other side is 135 m, with 7 auxiliary piers at this side span. Both towers are made of reinforced concrete. This bridge's main span utilizes an LSOB main girder construction, with the load-bearing section being a steel structure on the lower side and a concrete deck slab on the upper side. The lower side's steel construction is mostly made up of the winding nozzle closed box girder and the I-girder longitudinal girders that flank it. This arrangement lowers fabrication costs while ensuring the ideal girder stiffness. Figure 2 shows the sectional features of the main girder.

The wind tunnel test was conducted in the second test section of the XNJD-1 wind tunnel, which possesses a cross-sectional size of 2.4 m (width) $\times$ 2.0 m (height) $\times$ 16.0 m (length). In the wind tunnel test, a scale-down ratio of 1:55

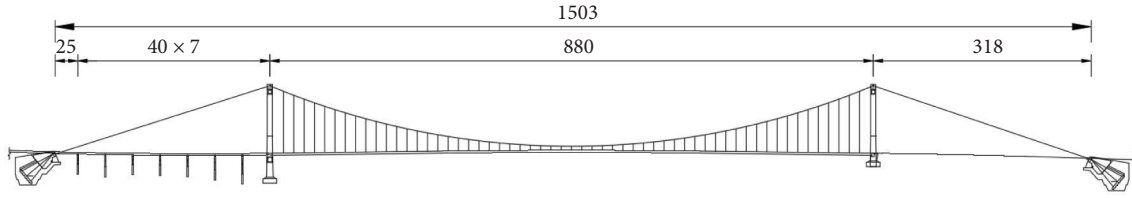


FIGURE 1: General arrangement of bridge (unit: m).

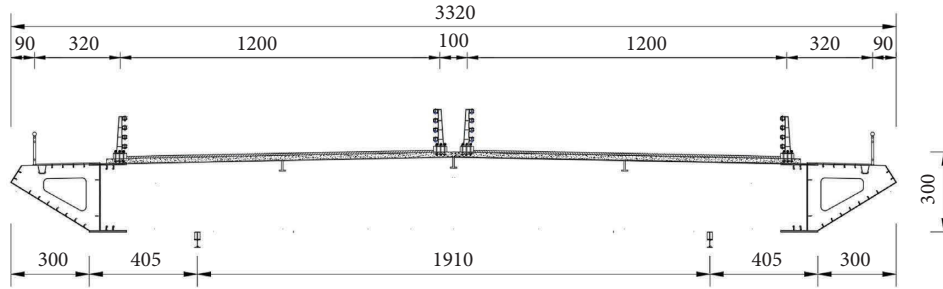


FIGURE 2: Cross-sectional configuration of the prototype LSOB girder (unit: cm).

was chosen for the test model, resulting in section model dimensions of 2.095 m (length), 0.604 m (width), and 0.055 m (height), with a blocking degree of less than 5%. The turbulence intensity is less than 0.5% in smooth flow, and the adjustable wind speed range of the test section is 1.0–45.0 m/s. With respect to the evaluation of damping ratio for steel-concrete composite bridges, the Design Manual for Roads and Bridges [23] recommends that the damping ratio should be 0.48%, so the vertical damping ratio for the test was 0.36%, while the torsional damping ratio was 0.34%. FRP plates were used to construct the longitudinal through-length I-girders on both sides of the girder, with the upper surface of the model being skinned and the railings made of engineering plastic plates. Eight tension springs suspended the section model on a bracket outside the cavity wall, ensuring that the girder could undergo vertical bending and torsional vibration. Rest assured, detailed information regarding the section model and the test parameters can be found in Figure 3 and Table 1, respectively. In the test, a set of eight extension springs were used to suspend a set of bracket system, and then the segmental model was connected to the bracket, and the vertical frequency of the segmental model system was controlled by designing the special spring stiffness. The designed mass blocks were mounted on the bracket system under the specified mass, and the torsional frequency and mass moment of inertia of the segmental model system were adjusted by adjusting the position of the mass blocks. During the wind tunnel of the adjusted segmental model system, the vibration response of the segmental model in the test was tested by a noncontact laser displacement sensor (ILD1401-200) installed on the bracket system, as shown in Figure 3(c).

The VIV test was conducted in a wind tunnel under a uniform flow field with wind speeds ranging from 0.5 to 6.5 m/s, corresponding to wind speeds ranging from 1.7 to 22.1 m/s on the actual bridge, in which the experimental

Reynolds number varied in the range of 1.9×10^3 – 2.4×10^4 . The vertical and torsional VIVs of the main girder were tested at angles of attack of 0° , $\pm 3^\circ$, respectively. Considering the characteristics of the wind in mountainous area, the attack angles selected to test are extended to $\pm 5^\circ$. The results are depicted in Figure 4, where y/d is the dimensionless vertical amplitude, y is the calculated vertical amplitude, d is the depth of the testing section, U/fd is the reduced wind speed, f_h is the vertical frequency of the suspension system, and f_a is the torsional frequency of the suspension system. These indicate that the original LSOB girder exhibits vertical and torsional VIV patterns under each attack angle, with a notably significant VIV response. The maximum amplitude values are observed at a attack angle of $+5^\circ$, where the maximum vertical amplitude (y/d) is 0.190, and the maximum torsional amplitude is 1.19° .

3. Numerical Simulation Study of VIV Triggering Mechanism

3.1. Calculation Method and Accuracy Verification. The flowchart depicting the cross-sectional flow–solid coupling analysis method using dynamic mesh is illustrated in Figure 5. As depicted, the initial step involves calculating the flow around the section of the girder in a stationary scenario. Subsequently, as the flow field reaches relative stability, the rigid body is subjected to motion under the influence of aerodynamic forces. This motion leads to further interaction between the fluid and solid components, thus initiating fluid–solid coupling. The rigid-body region’s motion rules are defined by compiling user-defined functions (UDFs) programmed in C. The dynamic mesh updating method employs the spring-based-smoothing (SBS) model. Forces and moments acting on the girder section within the flow field are obtained from the results. Subsequently, displacement, acceleration, and velocity at the next time step are determined

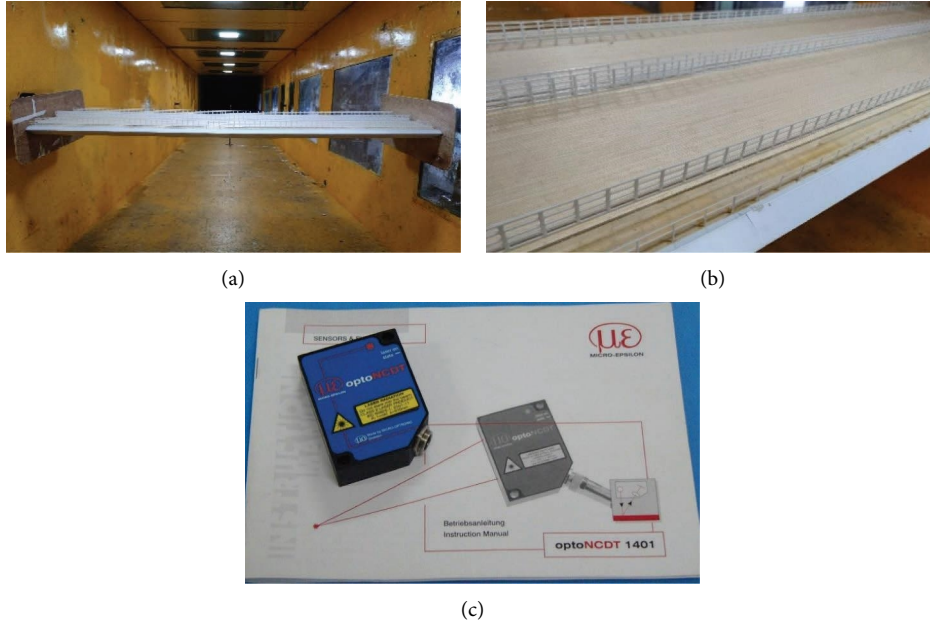


FIGURE 3: The section model configuration details in wind tunnel. (a) Section model overall. (b) Detail of bridge deck ancillary elements. (c) Noncontact laser displacement sensor (ILD1401-200).

TABLE 1: Segment model test parameters.

Properties	Unit	Model
Mass per unit length	kg/m	9.653
Mass moment of inertia per unit length	Kg-m ² /m	0.301
Vertical frequency	Hz	2.625
Torsional frequency	Hz	5.932

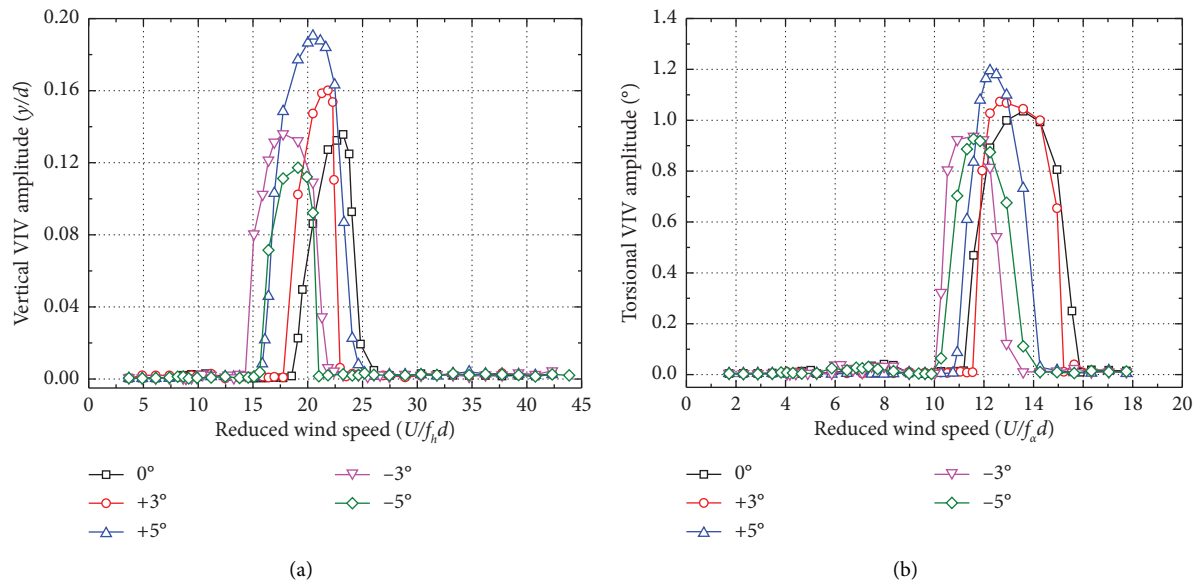


FIGURE 4: VIV response of the original LSOB girder. (a) Vertical. (b) Torsional.

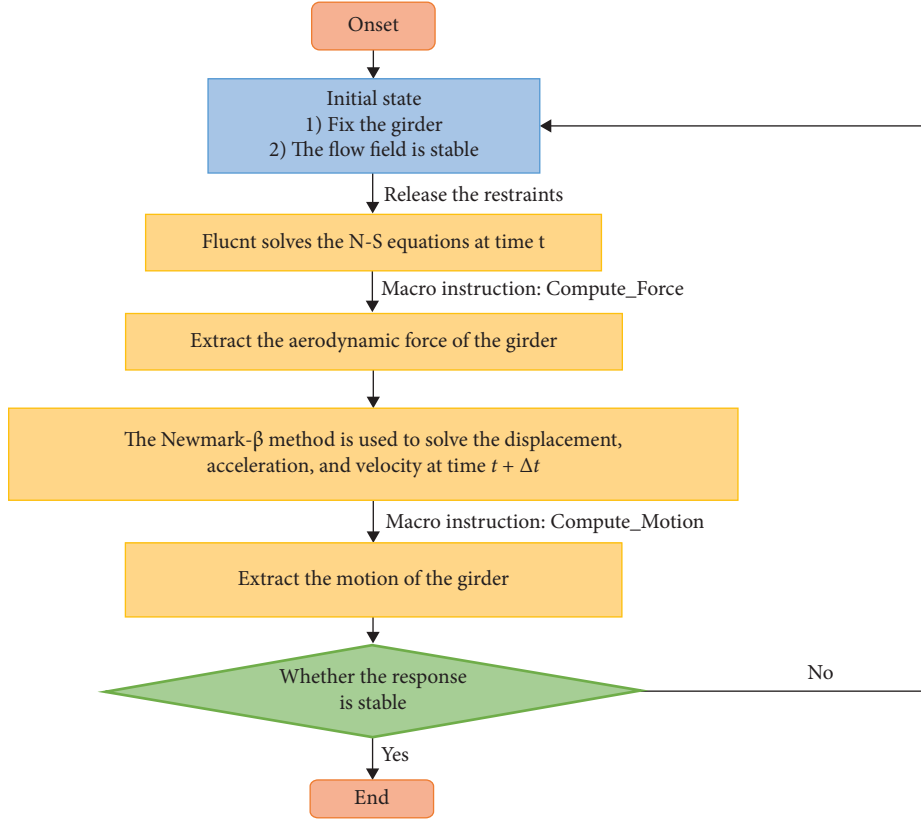


FIGURE 5: Flowchart for numerical calculation of vortex excitation vibration.

utilizing the Newmark- β method, a numerical integration technique in dynamics. The motion of the rigid body is then defined by the macro DEFINE.CG.MOTIONG, ensuring the corresponding update of the rigid body's position.

The solution to the vortex-excited vibration response is grounded in a fluid-solid coupling approach, which entails solving the hydrodynamic and structural equations of motion sequentially at each time step by integrating the Newmark- β method into the UDFs. This method was used to solve many other similar problems [7, 24]. The equations governing the vertical and torsional motion of the structure can be expressed as follows:

$$\begin{aligned} m[h''(t) + 2\xi_h\omega_h h'(t) + \omega_h^2 h(t)] &= F(t), \\ I_m[\alpha''(t) + 2\xi_\alpha\omega_\alpha \alpha'(t) + \omega_\alpha^2 \alpha(t)] &= M(t), \end{aligned} \quad (1)$$

where m is the mass per unit length of the girder section, I_m is the mass per unit length moment of inertia, and $F(t)$ and $M(t)$ are the vertical load and the moment of the center of mass of the girder, respectively. $h''(t)$, $h'(t)$, and $h(t)$ denote the acceleration, velocity, and displacement of the vertical motion of the girder at time t , respectively. $\alpha''(t)$, $\alpha'(t)$, and $\alpha(t)$ denote the acceleration, velocity, and displacement of the torsional motion of the girder at time t , respectively. ω_h denotes the structure's vertical self-oscillation frequency, and the structure's vertical self-oscillation frequency, and h

and α are the vertical damping ratio and torsional damping ratio, respectively.

The Newmark- β method is based on a stepwise integration format, assumed over t to $t + \Delta t$

$$\begin{aligned} x'_{t+\Delta t} &= x'_t + [(1 - \delta)x''_t + \delta x''_{t+\Delta t}]\Delta t, \\ x_{t+\Delta t} &= x_t + x'_t\Delta t + \left[\left(\frac{1}{2} - \beta\right)x''_t + \beta x''_{t+\Delta t}\right]\Delta t^2, \end{aligned} \quad (2)$$

where δ and β are parameters adjusted according to the requirements of integration accuracy and stability, resulting in the use of $x_{t+\Delta t}$, x_t , x'_t , and x''_t . The expression is expressed as

$$\begin{aligned} x''_{t+\Delta t} &= \frac{1}{\beta\Delta t^2}(x_{t+\Delta t} - x_t) - \frac{1}{\beta\Delta t}x'_t - \left(\frac{1}{2\beta} - 1\right)x''_t, \\ x'_{t+\Delta t} &= \frac{\delta}{\beta\Delta t}(x_{t+\Delta t} - x_t) + \left(1 - \frac{\delta}{\beta}\right)x'_t + \left(1 - \frac{\delta}{2\beta}\right)x''_t\Delta t. \end{aligned} \quad (3)$$

Reassociate the $t + \Delta t$ momentary equations of motion

$$mx''_{t+\Delta t} + cx'_{t+\Delta t} + kx_{t+\Delta t} = p_{t+\Delta t}, \quad (4)$$

can eventually be obtained, thus finding the response $x''_t\Delta t$ and $x'_t\Delta t$

$$\left(k + \frac{1}{\beta\Delta t^2}m + \frac{\delta}{\beta\Delta t}c\right)x_{t+\Delta t} = p_{t+\Delta t} + m\left[\frac{1}{\beta\Delta t^2}x_t + \frac{1}{\beta\Delta t}x'_t + \left(\frac{1}{2\beta} - 1\right)x''_t\right] + c\left[\frac{\delta}{\beta\Delta t}x_t + \left(\frac{\delta}{2\beta} - 1\right)x'_t\right]. \quad (5)$$

The Newmark- β method is calculated as follows:

- (1) Know k , m , and c
- (2) Given the initial conditions (0) , $x''(0)$, and $x'(0)$
- (3) Select the time step Δt as well as the parameters δ and β and calculate the integration constant
- (4) Formation of effective stiffness
- (5) Calculate the effective load at the moment $t + \Delta t$
- (6) Solve for the displacement at the moment $t + \Delta t$
- (7) Find the acceleration and velocity at the moment $t + \Delta t$

The Newmark- β method uses $t + \Delta t$ moment equilibrium conditions and is an implicit integration method, which is unconditionally stable when $\delta \geq 1/2$ and $\beta \geq 1/4(1/2 + \delta)^2$, and the size of the time step Δt affects the solution accuracy, but not the stability. When $\delta = 1/4$, $\beta = 1/2$, it is the constant average acceleration method, assuming that the acceleration is constant from t to the moment $t + \Delta t$, which is taken to be constant $1/2(x''_t + x''_{t+\Delta t})$. This method is used in the UDF secondary development procedure in this paper.

To ensure the comprehensiveness of our simulations, we have set the convergence term residual at 0.00001 and designated the wall as nonsliding. The SST $k - \omega$ turbulence model is adopted [13, 25]. The flow inlet extends from the velocity inlet of the left boundary to the pressure outlet of the right boundary, ensuring a thorough representation of the fluid dynamics. Given that the VIV response of the LSOB girder was most pronounced at a $+5^\circ$ attack angle in prior wind tunnel tests, subsequent numerical simulations were conducted under this condition. Both the computational domain and boundary settings are illustrated in Figure 6, while the distribution of mesh details is depicted in Figure 7.

The vertical vortex response of the original LSOB girder, obtained through numerical simulation, is depicted in Figure 8. The VIV wind speed interval, the starting wind speed, and their corresponding amplitudes show good agreement with the experimental results of the sectional model. The observed agreement between the experimental and calculated values provides support for the validity of the flow field properties obtained from the numerical simulation. As such, the CFD method could be used to efficiently analyze the LSOB girder's VIV triggering mechanism and flow field characteristics.

3.2. Research on Flow Field Characteristics and VIV Triggering Mechanism. The dimensionless vertical displacement and lift coefficient of the original LSOB girder for the starting wind speed $U/f_h d = 15.4$ are plotted as follows (Figure 9). The phase difference between the lift coefficient and displacement is only 0.039π , which indicates that the pneumatic lift has done positive work.

Figure 10 illustrates the flow and pressure distribution of the LSOB girder during one vibration cycle. This analysis was conducted at a significant vertical VIV starting wind speed of $U/f_h d = 15.4$. The results show that large-scale vortices coexist on both the upper and lower surfaces of the bridge deck. As the girder moves upward, the lift force of the main girder peaks, primarily due to the negative pressure induced by the motion of vortices T0, X1, and X2. The negative pressure increases due to the reduction in size and shedding of the vortex on the upper side, along with the negative pressure generated by vortex Y2 at the sloping web of the wind nozzle on the leeward side. This leads to a decrease in the pressure difference between the upper and lower surfaces of the girder, ultimately resulting in the minimum lift of the girder section. Vortex Y2 is completely dislodged, and X2 and X3-1 merge to form T0, initiating the next cycle.

As demonstrated by the research shown above, two primary types of vortices induce the VIV of the LSOB girder. The first type is the vortex created at the wind fairing's sloping web on the leeward side. The other kind is the vortex formed at the girder's upstream top surface. These vortices interact with one another to increase the girder's vibration, which leads to the formation of a classic Kamen vortex street in the wake area.

4. Influence of Lower Stabilising Plate on VIV Response and Flow Field Characteristics of LSOB Girders

The lower stabilizing plate is a commonly employed pneumatic measure for VIV suppression in various girder types. In the case of LSOB girders, a lower central stabilizing plate with a length of 340 cm is introduced in this study to explore its impact on the VIV performance of the girders. The section following the installation of the stabilizing plate is referred to as the LSOB-LVP section, as depicted in Figure 11.

The VIV response of the LSOB-LVP girder was obtained under the same wind tunnel test parameters as described in the previous section, and the corresponding test results are presented in Figure 12. Obvious vertical and torsional VIV intervals are observed in the girder at all wind angles of attack. However, the vortex amplitudes of the LSOB-LVP girder at each angle of attack are significantly reduced compared to those of the original LSOB girder, with a reduction of more than 40%. The results suggest that the lower central stabilizing plate has a more pronounced effect on enhancing the VIV performance of the LSOB section.

The flow field of the LSOB-LVP section under a vertical VIV starting wind speed of $U/f_h d = 17.1$ was analyzed using the same dynamic grid numerical simulation technique employed in the previous section, as illustrated in Figure 13. The simulation results indicate that the installation of the

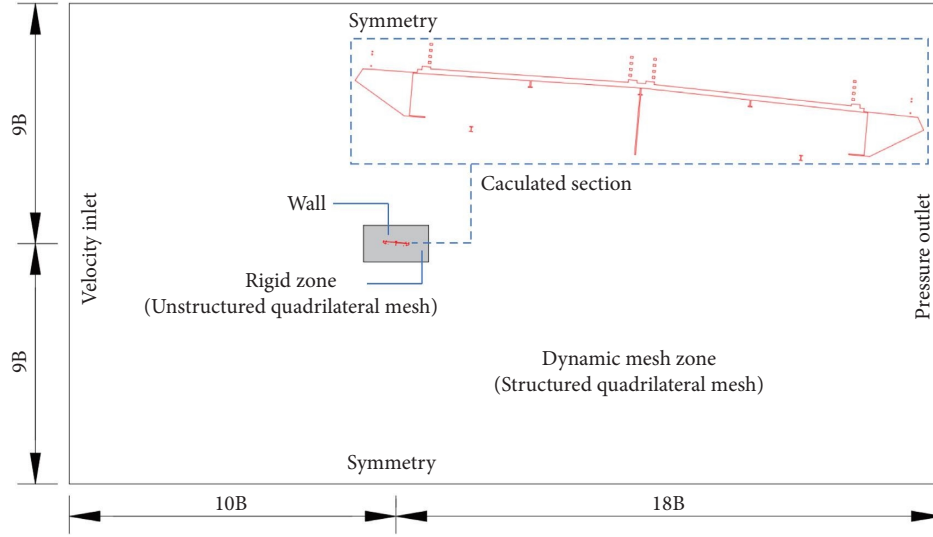


FIGURE 6: Computational domain and boundary settings.

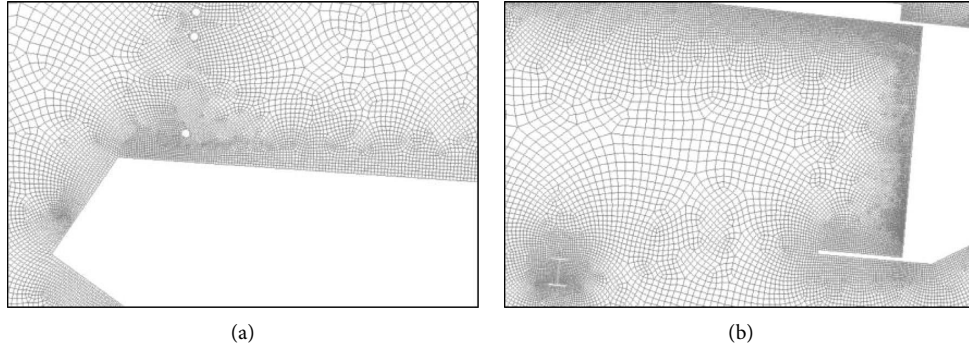


FIGURE 7: Distribution of mesh details. (a) The grid near the fairing and handrail. (b) The grid near the edge girder.

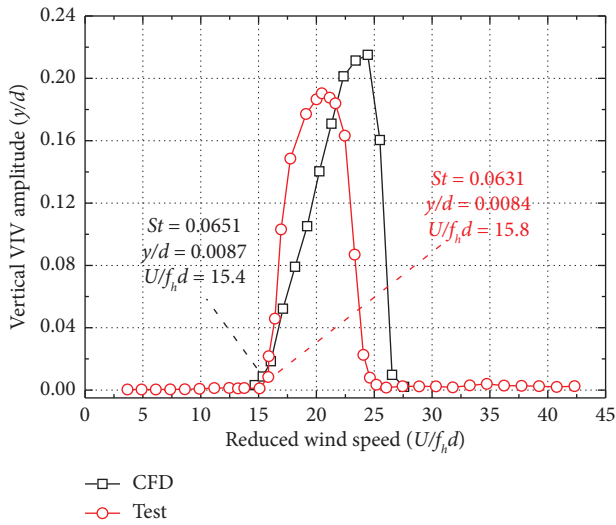


FIGURE 8: Numerical simulation and experimental vertical VIV responses of the LSOB girder.

lower stabilizer plate facilitates the comprehensive redistribution of vortices beneath the girder, representing a favorable alteration. Compared to the vortex Y1 in the

original cross-section (the height of Y1 is 0.82 times the height of the beam and the width is 0.48 times the width of the beam), the maximum vortex Y0 on the lower side of the LSOB-LVP beam is only 0.73 times the height of the beam and 0.22 times the width of the beam, a significant reduction in the vortex dimensions has occurred. Consequently, the vortices beneath the girder transform into smaller vortices, with most not shedding. This transformation reduces the intensity of vortex shedding in the wake region, thereby enhancing the VIV performance of the LSOB girder.

Notably, a small vortex Y0 appears at the lower end of the lower stabilizing plate, detached and operating at a higher frequency than the detachment frequency of the primary vortex in the wake flow. Despite quickly merging with the vortex downstream after separation, this vortex does not significantly affect the flow field in the wake region. Nonetheless, further investigation is warranted to assess the impact on the girder when shedding occurs at the stabilizing plate.

To investigate whether Y0 vortex shedding triggers vortex-excited vibration, the flow field state of the girder at rest before vortex-excited vibration occurs is simulated by immobilizing the LSOB-LVP girder. The static wind state of

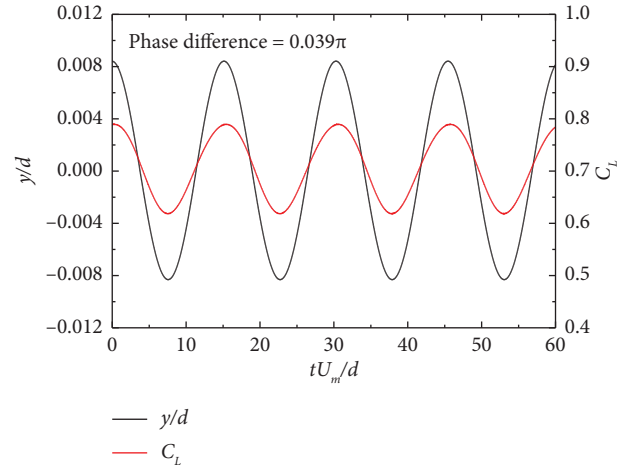


FIGURE 9: Plot of dimensionless vertical displacement of an LSOB girder at $U/f_h d = 15.4$ versus C_L time course.

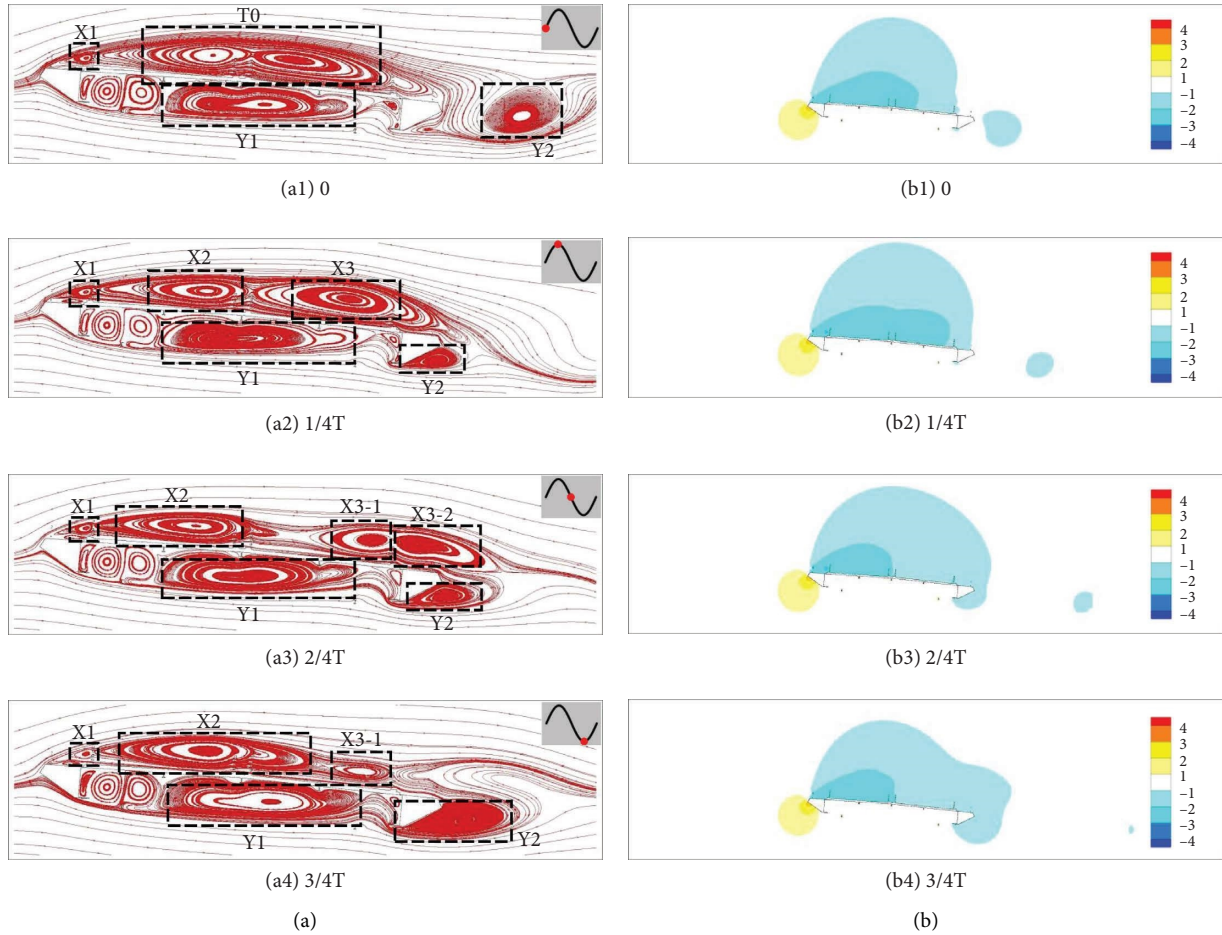
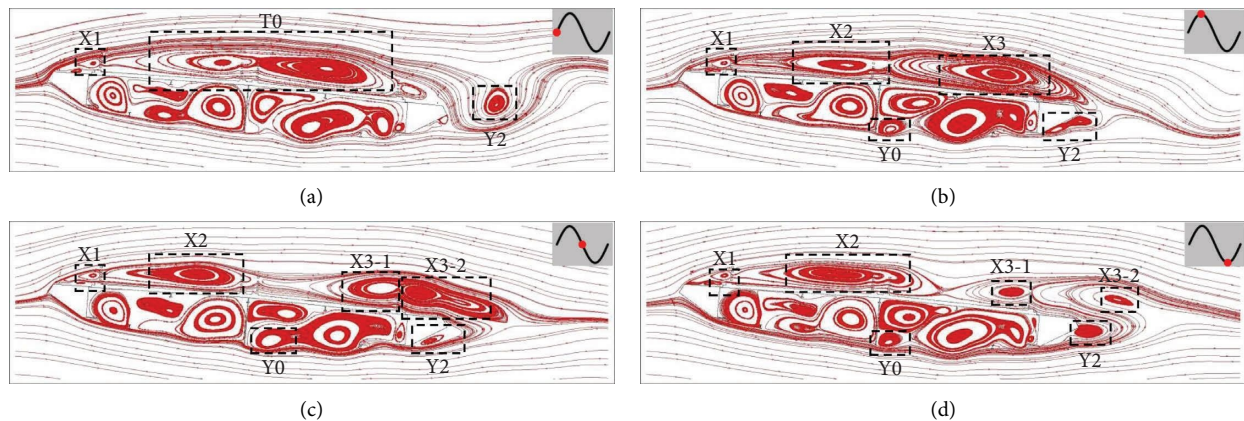
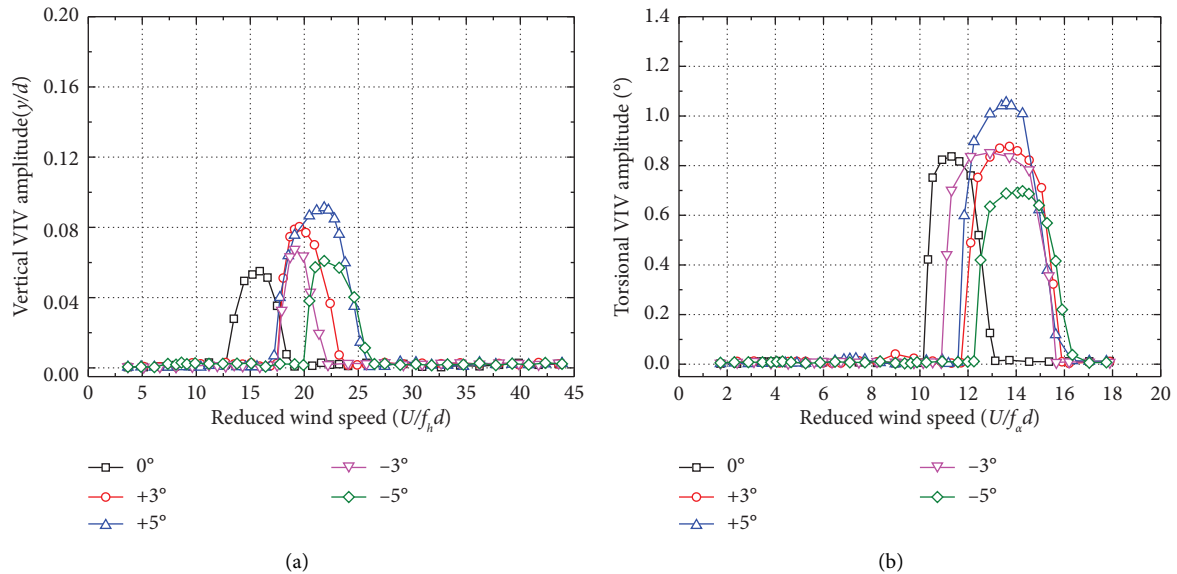
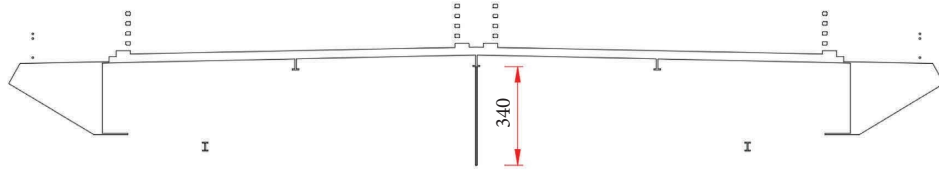


FIGURE 10: Flow field characteristics of an LSOB girder under $U/f_h d = 15.4$ during one vibration cycle. (a) Streamline diagram. (b) Pressure distribution diagram.

the LSOB-LVP girder is illustrated in Figure 14. It is observed that after immobilizing the girder, there are no significant changes in the vortices on the upper surface or in the wake region of the lower surface, except for vortex Y0.

Vortex Y0 is not generated when the LSOB-LVP girder is at rest. By extracting the section lift coefficients for the stationary and VIV states, as depicted in Figure 15, the variation in the lift coefficient of the LSOB-LVP girder in the



vibration state reveals two notable frequencies, namely, 2.657 Hz and 5.314 Hz. In contrast, the variation in the lift coefficient in the stationary state exhibits only one notable frequency, 2.658 Hz. This finding confirms that the lift fluctuation of higher frequency but lower energy, caused by the shedding vortex Y0, occurs after the onset of VIV. Thus, the vortex is not a trigger for LSOB-LVP girder vortex-excited vibration.

5. Study on the Role of VIV Performance Improvement of Rail Deflectors for Inspection Vehicles

Commonly used in the inspection vehicles, double-side guide vanes serve as an aerodynamic solution for VIV suppression in box girders. Vortex shedding still happens on the leeward side of the LSOB girder, even though the

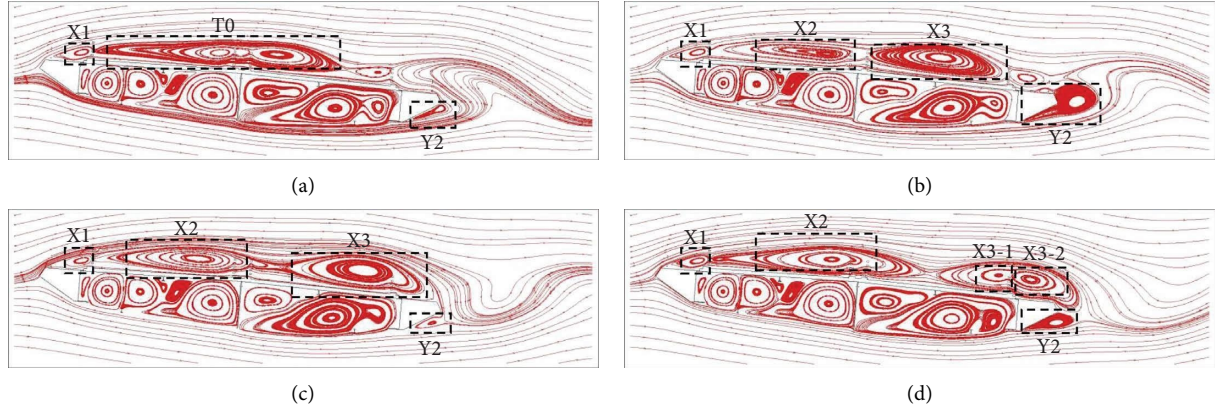


FIGURE 14: Static section flow diagram of LSOB-LVP girder under $U/f_h d = 17.1$. (a) 0. (b) $1/4T$. (c) $2/4T$. (d) $3/4T$.

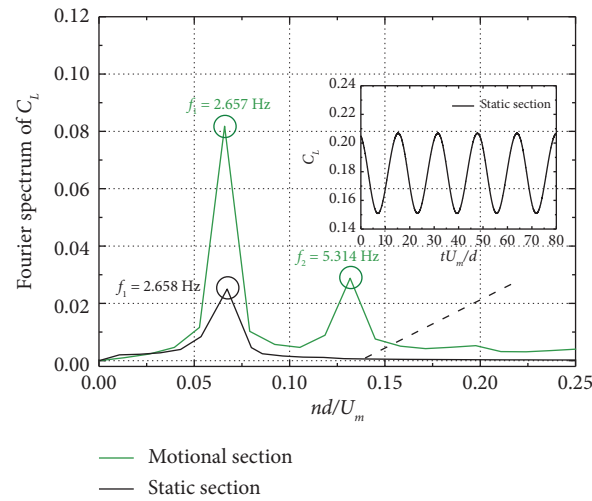


FIGURE 15: The lift coefficient spectrogram of the LSOB-LVP girder.

lower side of the section's flow field has significantly improved after the lower stabilizing plate was installed. So this section explores the effect of double-side guide vanes on the VIV performance of the LSOB girder by forming the LSOB-GV section. The guide vanes, with an inclination angle of 17° , are installed at the gantry rail, as illustrated in Figure 16.

The VIV amplitudes of different LSOB girder sections at each attack angle are depicted in Figure 17. Comparison of the VIV response between the LSOB-LVP and LSOB-GV girders reveals that the VIV response of the girders at each attack angle decreases after installing the guide vane. However, the decrease, although not substantial, remains within 30% compared to the value of the LSOB-LVP section. This change can be attributed to examining the winding state of the LSOB-GV girder. The main disparity between the LSOB-GV girder and the flow diagram of the LSOB-LVP section is the significant reduction in the size of wake vortex Y2 at the sloping web on the leeward side, as illustrated in Figure 18. Versus the LSOB-LVP section, the width of the vortex Y2 decreased by 67% and the height by 83% after the addition of the double-side guide vanes. This observation

suggests that while installing a guide vane at the gantry rail can improve the VIV performance of the LSOB girder to some extent, it cannot be entirely eliminated due to the persistence of significant vortex shedding on the upper side of the girder, coupled with the presence of minor wake vortex shedding on the lower side.

Above all, it is found that the VIV performance optimization of the LSOB girder can be carried out by improving the flow field at the lower semi-open region of the girder, which is effectively demonstrated by the effect of the lower stabilizing plate and guide vanes. This observation can provide design guidance for aerodynamic countermeasures for LSOB girders. However, it is worth mentioning that if the vortex-excited vibration of the LSOB girder is to be completely or significantly suppressed, it is not enough to merely improve the flow field on the lower side of the girder. Additional aerodynamic measures need to be proposed for the improvement of the upper side flowing bypassing state. Since this paper is a preliminary investigation on the VIV performance of the LSOB girder, a new type of girder, it does not involve the study of the vibration suppression effect of more commonly used aerodynamic measures. A more

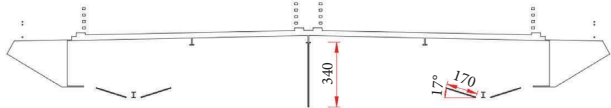
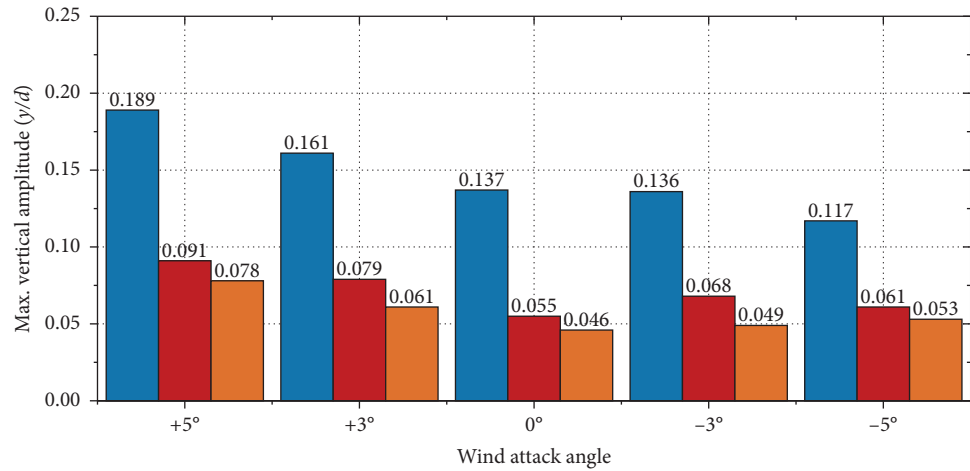
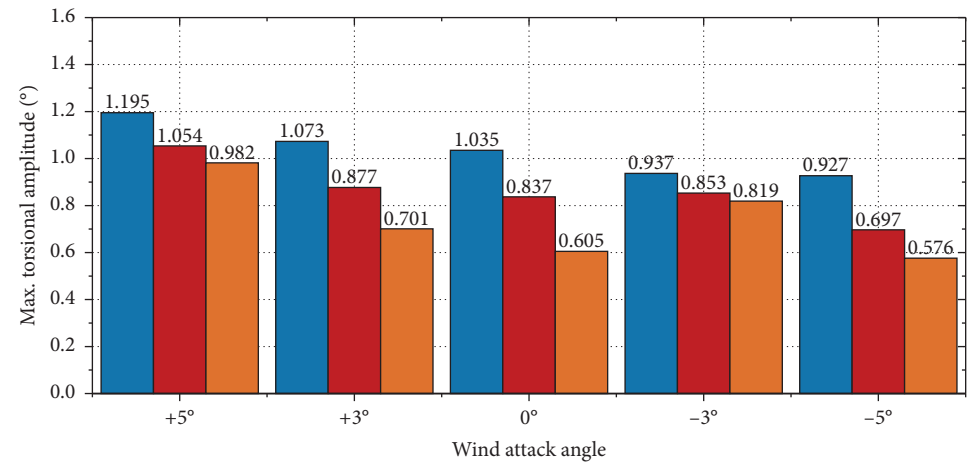


FIGURE 16: Cross-sectional configuration of the LSOB-GV girder (unit: cm).



■ The original LSOB section
■ The LSOB-LVP section
■ The LSOB-GV section

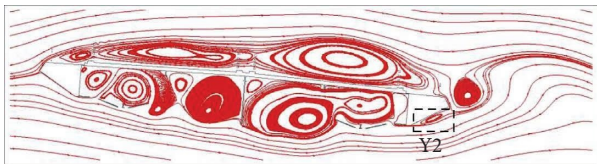
(a)



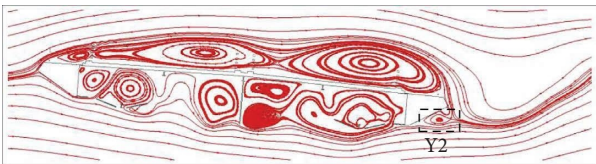
■ The original LSOB section
■ The LSOB-LVP section
■ The LSOB-GV section

(b)

FIGURE 17: VIV response of different LSOB girder sections. (a) Vertical. (b) Torsional.



(a)



(b)

FIGURE 18: Continued.

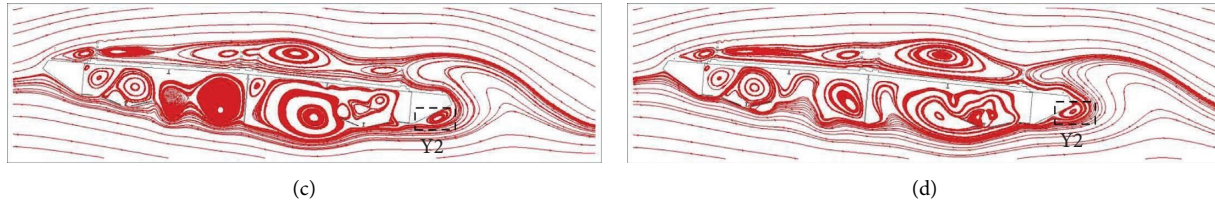


FIGURE 18: Static section flow diagram of LSOB-GV girders. (a) 0. (b) 1/4T. (c) 2/4T. (d) 3/4T.

systematic and in-depth study on the VIV control aerodynamic measures of LSOB girders will be carried out subsequently.

6. Conclusions

Based on the results of wind tunnel tests and numerical simulations of the section model involved in this paper, the main conclusions are drawn as follows:

- Under a damping ratio of about 0.35%, there is a significant VIV phenomenon in the LSOB girder. The installation of the lower central stabilizing plate can play a role in reducing the VIV response of the LSOB girder at all wind angles of attack, with a vertical amplitude reduction in the range of 40%–50% and a torsional amplitude reduction in the range of 10%–25%. The VIV performance of the LSOB girder can be further optimized by installing double-side guide vanes at the gantry rail, but the effect is limited.
- The CFD simulation results show that the vortex-excited vibration of the LSOB girder is mainly caused by two types of vortices, one at the inclined web of the wind nozzle on the leeward side, and the other at the upstream upper surface of the girder. The interaction of these vortices contributes to the vortex-excited vibration of the girder and eventually forms a typical Kamen vortex street in the wake region.
- For the VIV of the LSOB girder in this paper, the practical vibration suppression ability of the lower central stabilizer plate is mainly attributed to its ability to facilitate the overall redistribution of vortices under the girder, which can change the large vortices under the original section into a series of smaller vortices, and most of which do not fall off, thus reducing the intensity of vortex shedding in the wake area and improving the VIV performance of the LSOB girder. The measurement reduces the intensity of vortex shedding in the wake region and improves the VIV performance of the LSOB. The guide vane's role is to further weaken the wake vortices at the leeward side of the sloping web.

It is worth mentioning that a more systematic study is needed for the VIV performance of LSOB beams, and the research results in this paper are only a preliminary exploration, which should be followed up by a systematic study for LSOB beams with different shape parameters, so as to obtain more general conclusions.

Data Availability Statement

All data used during the study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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